Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A survey was conducted on students' preferred modes of transportation to school. The results are summarized in the two-way frequency table below. \\ | | Bus | Bicycle | Walking | Total | |-----------------|-----|---------|---------|-------| | Under 16 | 45 | 30 | 25 | 100 | | 16 and Over | 60 | 15 | 25 | 100 | | Total | 105 | 45 | 50 | 200 | Let B be the event that a randomly selected student prefers to travel by Bus. Let U be the event that a randomly selected student is Under 16 years old. \\ What is the probability that a student prefers to travel by Bus, given that they are Under 16 years old? Mathematically, this is denoted as $P(B|U)$.

We are asked to find the conditional probability $P(B|U)$, which represents the probability that a student prefers to travel by Bus given that they are Under 16 years old. \\ The formula for conditional probability is $P(B|U) = \frac{P(B \cap U)}{P(U)}$. \\ From the table: - The number of students who prefer Bus AND are Under 16 ($B \cap U$) is 45. \\ - The total number of students who are Under 16 (U) is 100. \\ Therefore, $P(B \cap U) = \frac{45}{200}$ and $P(U) = \frac{100}{200}$. \\ Substituting these values.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A civil engineer measures the width of a river BC by setting up parallel stakes such that $DE \parallel BC$. Point D is on line AB and E is on line AC. If $AD = 45$ m, $DB = 75$ m, and $DE = 36$ m, what is the width of the river BC?

First compute the total line length: $AB = AD + DB = 45 + 75 = 120$ m. Because $DE \parallel BC$, $\Delta ADE \sim \Delta ABC$. The proportion of corresponding sides is $\frac{DE}{BC} = \frac{AD}{AB} \Rightarrow \frac{36}{BC} = \frac{45}{120} = \frac{3}{8}$. Cross-multiplying: $3 \cdot BC = 36 \times 8 = 288 \Rightarrow BC = 96$ m.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A 5.5-foot-tall surveyor casts a 4.4-foot shadow on level ground. At the exact same moment, a nearby radio antenna tower casts a 36-foot shadow. How tall is the antenna tower?

Sunlight rays strike level ground at the same angle of elevation, creating similar right triangles between objects and their shadows: $\frac{\text{Height}}{\text{Shadow}} = \frac{5.5}{4.4} = 1.25$. For the tower: $\frac{H}{36} = 1.25 \Rightarrow H = 36 \times 1.25 = 45$ ft.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC, segment $DE \parallel BC$. The segment lengths are $AD = x$, $DB = x + 3$, $AE = x + 2$, and $EC = x + 7$. What is the value of x?

By the Triangle Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{x}{x+3} = \frac{x+2}{x+7}$. Cross-multiplying: $x(x + 7) = (x + 3)(x + 2) \Rightarrow x^2 + 7x = x^2 + 5x + 6$. Subtracting x^2 from both sides gives $7x = 5x + 6 \Rightarrow 2x = 6 \Rightarrow x = 3$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC, ray AD bisects $\angle BAC$ with D lying on side BC. If $AB = x + 3$, $AC = 2x$, $BD = 4$, and $DC = 6$, what is the value of x?

By the Angle Bisector Theorem, the ratio of the divided base segments equals the ratio of the adjacent sides: $\frac{BD}{DC} = \frac{AB}{AC} \Rightarrow \frac{4}{6} = \frac{x+3}{2x}$. Cross-multiplying: $4(2x) = 6(x + 3) \Rightarrow 8x = 6x + 18 \Rightarrow 2x = 18 \Rightarrow x = 9$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In an hourglass figure where $AB \parallel CD$ with intersection at E, $AE = 2x + 1$, $EC = 5x - 4$, $BE = 6$, and $ED = 9$. What is the value of x?

By similarity $\Delta ABE \sim \Delta CDE$: $\frac{AE}{EC} = \frac{BE}{ED} \Rightarrow \frac{2x+1}{5x-4} = \frac{6}{9} = \frac{2}{3}$. Cross-multiplying:

$3(2x + 1) = 2(5x - 4) \Rightarrow 6x + 3 = 10x - 8 \Rightarrow 4x = 11 \Rightarrow x = \frac{11}{4} = 2.75$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In right ΔABC with altitude $CD \perp AB$, $AD = 2$ and $DB = 8$. What is the ratio of the legs $\frac{AC}{BC}$?

By the Geometric Mean Leg Theorems: $AC^2 = AB \cdot AD$ and $BC^2 = AB \cdot DB$. Dividing the two equations: $\frac{AC^2}{BC^2} = \frac{AD}{DB} = \frac{2}{8} = \frac{1}{4}$. Taking the square root gives $\frac{AC}{BC} = \sqrt{\frac{1}{4}} = \frac{1}{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two similar triangles have areas of $45\ \text{cm}^2$ and $80\ \text{cm}^2$. If the perimeter of the smaller triangle is $27\ \text{cm}$, what is the perimeter of the larger triangle?

The ratio of the areas is $\frac{A_2}{A_1} = \frac{80}{45} = \frac{16}{9} = k^2$. The linear scale factor is $k = \sqrt{\frac{16}{9}} = \frac{4}{3}$. Perimeters scale linearly by k: $P_2 = 27 \times \frac{4}{3} = 36\ \text{cm}$.

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9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The point P has coordinates $(-3, 4)$. Point P' is the image of P after a reflection across the line $y = x$. What are the coordinates of P'?

When a point (x, y) is reflected across the line $y = x$, its image has coordinates (y, x). Given the point P has coordinates $(-3, 4)$. Applying the reflection rule, the coordinates of P' will be $(4, -3)$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In right ΔABC with right angle at C and altitude $CD \perp AB$, leg $BC = 12$ and hypotenuse $AB = 16$. What is the length of segment DB?

By the Geometric Mean Leg Theorem, $BC^2 = AB \cdot DB$. Substituting the known lengths: $12^2 = 16 \cdot DB \Rightarrow 144 = 16 \cdot DB \Rightarrow DB = \frac{144}{16} = 9$. (Consequently, $AD = 16 - 9 = 7$).

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 5x + 8$ in volume of prisms and cylinders, evaluate $f(5)$.

Final Answer: 33. Substitute $x = 5$: $f(5) = 5(5) + 8 = 33$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 9$ in volume of pyramids and cones, evaluate $f(2)$.

Final Answer: 21. Substitute $x = 2$: $f(2) = 6(2) + 9 = 21$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for volume of pyramids and cones consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · AdvancedIn the hard Module 2, questions 18–22 are typically high-difficulty multi-step problems with parameters k. If a student has 7 minutes remaining for the last 3 questions, which approach optimizes score yield: attempting all 3 partially, or using Desmos regression and choice elimination to guarantee complete accuracy on 2 of them?

Optimal Strategy: On the Digital SAT's Item Response Theory (IRT) scoring model, fully correct answers yield substantially higher raw theta weights than rushed guesses. Allocating 3.5 minutes each to 2 questions using Desmos graphing/regression guarantees 2 correct hard items (+40–60 scaled points), while placing a rapid educated guess on the 3rd question preserves time with zero guessing penalty.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Polygons & Quadrilaterals**

A circular archery target has three concentric rings of radii 10 cm, 20 cm, and 30 cm. If an arrow hits the target completely at random, what is the exact probability that it lands in the middle ring (between radii 10 cm and 20 cm)?

Step 1: Calculate the total area of the target: The outer circle has radius $R = 30$ cm: $\text{Area}_{\text{total}} = \pi(30^2) = 900\pi$ cm^2. Step 2: Calculate the area of the middle ring (annulus): The middle ring is bounded between $r_1 = 10$ cm and $r_2 = 20$ cm: $\text{Area}_{\text{ring}} = \pi(20^2) - \pi(10^2) = 400\pi - 100\pi = 300\pi$ cm^2. Step 3: Compute the geometric probability.

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Circles**

A circle is inscribed inside a square of side length 10 cm. A point is chosen at random inside the square. What is the exact probability that the point lands outside the circle (in the four shaded corners)?

Solution: The diameter of the inscribed circle equals the side of the square (10 cm), so the radius is $r = 5$ cm. Step 1 (Total area): $\text{Area(square)} = 10^2 = 100$ cm^2. Step 2 (Circle area): $\text{Area(circle)} = \pi(5^2) = 25\pi$ cm^2. Step 3 (Corner area): $\text{Area(outside)} = 100 - 25\pi$. Step 4 (Probability): $P(\text{outside}) = \frac{100 - 25\pi}{100} = 1 - \frac{\pi}{4} \approx 1 - 0.7854 = 0.2146$ (21.5%).

17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Area, Surface Area & Volume**

[Probability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Transformations & Symmetry**

A cone and a cylinder have identical base radii of 6 cm. If both solids have the same volume of 180π cm^3, find the height of the cylinder and the height of the cone.

Solution: The base area for both solids is $B = \pi r^2 = \pi(6^2) = 36\pi$ cm^2. For the cylinder: $V_{\text{cyl}} = B \cdot h_{\text{cyl}} \Rightarrow 180\pi = 36\pi \cdot h_{\text{cyl}} \Rightarrow h_{\text{cyl}} = \frac{180\pi}{36\pi} = 5$ cm. For the cone:

$V_{\text{cone}} = \frac{1}{3}B \cdot h_{\text{cone}} \Rightarrow 180\pi = \frac{1}{3}(36\pi) \cdot h_{\text{cone}} = 12\pi \cdot h_{\text{cone}} \Rightarrow h_{\text{cone}} = \frac{180\pi}{12\pi} = 15$ cm. .

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Foundations & Reasoning

A rhombus has diagonals of lengths 16 cm and 12 cm. Find the perimeter and area of the rhombus.

Step 1: Use diagonal properties of a rhombus: The diagonals of a rhombus are perpendicular bisectors of each other. They divide the rhombus into four congruent right triangles with legs of lengths $\frac{16}{2} = 8$ cm and $\frac{12}{2} = 6$ cm. Step 2: Find the side length s: Using the Pythagorean theorem: $s = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ cm. Step 3: Calculate the perimeter: $\text{Perimeter} = 4s = 4(10) = 40$ cm. Step 4: Calculate the area.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Coordinate Geometry

From an external point P, two tangents PA and PB are drawn to a circle with radius $r = 5$ cm. The distance from P to the center of the circle O is 13 cm. Find the length of the tangents PA and PB, and find the perimeter of quadrilateral $PAOB$.

Step 1: Recognize tangent-radius perpendicularity: Tangents are perpendicular to the radius at the point of tangency, so $\angle PAO = \angle PBO = 90^\circ$. ΔPAO is a right triangle with hypotenuse $PO = 13$ and leg $OA = 5$. Step 2: Solve for tangent length using the Pythagorean theorem: $PA = \sqrt{PO^2 - OA^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$ cm. By equal tangent lengths from an external point, $PB = PA = 12$ cm. Step 3: Compute the perimeter of $PAOB$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 6$ in cross sections of solids, evaluate $f(2)$.

a. 12 ✓ **b.** 15

c. 6 **d.** 16

Substitute $x = 2$: $f(2) = 3(2) + 6 = 12$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 7$ in solids of revolution, evaluate $f(2)$.

a. 15 ✓ **b.** 19

c. 8 **d.** 19

Substitute $x = 2$: $f(2) = 4(2) + 7 = 15$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for cross sections of solids consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solids of revolution consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for volume of prisms and cylinders consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for volume of spheres and hemispheres consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
For the model $f(x) = 2x + 3$ in volume of spheres and hemispheres, evaluate $f(3)$.

a. 9 ✓
b. 11
c. 6
d. 13

Substitute $x = 3$: $f(3) = 2(3) + 3 = 9$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
A data set recorded for volume of composite solids consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5
b. 23.5
c. 24.5 ✓
d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
For the model $f(x) = 3x + 4$ in volume of composite solids, evaluate $f(3)$.

a. 13 ✓
b. 16
c. 9
d. 17

Substitute $x = 3$: $f(3) = 3(3) + 4 = 13$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
A point is chosen at random on a segment of length 11 cm. Find the probability it lands within 2 cm of the left endpoint.

a. $\frac{1}{6}$
b. $\frac{2}{11}$ ✓
c. 1
d. $\frac{9}{11}$

Geometric probability compares lengths: $P = \frac{\text{favorable length}}{\text{total length}} = \frac{2}{11} = \frac{2}{11}$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A circular dartboard has radius 15 cm with a bullseye of radius 1 cm at the center. Assuming a dart lands at a uniformly random point on the board, find the probability it lands in the bullseye.

a. $\frac{1}{226}$

b. $\frac{1}{225}$ ✓

c. $\frac{224}{225}$

d. $\frac{1}{15}$

Since both areas are circles, π cancels: $P = \frac{\pi r^2}{\pi R^2} = \frac{1^2}{15^2} = \frac{1}{225} = \frac{1}{225}$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The probability that an event occurs is $\frac{11}{13}$. What is the probability that the event does NOT occur?

a. 0

b. $\frac{2}{13}$ ✓

c. 1

d. $\frac{11}{13}$

By the complement rule, $P(\text{not } A) = 1 - P(A) = 1 - \frac{11}{13} = \frac{2}{13}$, which is $\frac{2}{13}$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Independent events A and B have $P(A) = \frac{1}{5}$ and $P(B) = \frac{2}{5}$. Find $P(A \text{ or } B)$.

a. $\frac{3}{5}$

b. $\frac{14}{25}$

c. $\frac{2}{25}$

d. $\frac{13}{25}$ ✓

$P(A \text{ or } B) = P(A) + P(B) - P(A)P(B) = \frac{1}{5} + \frac{2}{5} - \frac{1}{5} \cdot \frac{2}{5} = \frac{13}{25}$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A circular target of radius 11 cm has a ring-shaped region between radius 3 cm and radius 9 cm. Find the probability a randomly landing dart hits this ring.

a. $\frac{9}{121}$

b. $\frac{81}{121}$

c. $\frac{72}{121}$ ✓

d. $\frac{36}{61}$

Ring area $= \pi(9^2 - 3^2)$, total area $= \pi(11^2)$. $P = \frac{9^2 - 3^2}{11^2} = \frac{72}{121} = \frac{72}{121}$.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two independent events A and B have $P(A) = \frac{1}{5}$ and $P(B) = \frac{6}{12}$. Find $P(A \text{ and } B)$.

a. $\frac{1}{10}$ ✓

b. $\frac{2}{11}$

c. $\frac{1}{5}$

d. $\frac{1}{2}$

For independent events, $P(A \text{ and } B) = P(A) \cdot P(B) = \frac{1}{5} \times \frac{6}{12} = \frac{1}{10}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A quality control laboratory receives components sorted into two bins: - **Bin A** contains 10 components, of which exactly 3 are defective and 7 are non-defective. - **Bin B** contains 10 components, of which exactly 6 are defective and 4 are non-defective. To conduct an inspection, an automated system first selects a bin at random: it rolls a fair standard six-sided die and selects **Bin A** if the result is 1 or 2; otherwise (if the result is 3, 4, 5, or 6), it selects **Bin B**. From the chosen bin, the system then randomly samples 2 components without replacement. If both sampled components are found to be defective, what is the probability that the components were drawn from **Bin A**?

a. $\frac{1}{11}$ ✓

b. $\frac{1}{9}$

c. $\frac{1}{6}$

d. $\frac{10}{11}$

We use Bayes' theorem combined with combinations to find the conditional probability $P(\text{Bin A} \mid \text{both defective})$. **Step 1: Determine prior probabilities of selecting each bin.** The die has 6 equally likely outcomes: - Bin A is selected if the die shows 1 or 2: $P(A) = \frac{2}{6} = \frac{1}{3}$ - Bin B is selected if the die shows 3, 4, 5, or 6: $P(B) = \frac{4}{6} = \frac{2}{3}$ **Step 2: Determine conditional probabilities of sampling 2 defective components without replacement.** Let.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A random sample of 200 high school students was surveyed regarding their grade level and whether they participate in school sports. The results are summarized in the two-way frequency table below: | | Participates in Sports (S) | Does Not Participate in Sports | Total | | :--- | :---: | :---: | :---: | | Grade 10 (T) | 48 | 32 | 80 | | Grade 11 | 42 | 78 | 120 | | Total | 90 | 110 | 200 | A student is selected at random from this sample. Let S be the event that the student participates in sports, and let T be the event that the student is in Grade 10. Which of the following statements is correct regarding the conditional probability $P(S \mid T)$ and the statistical independence of events S and T?

a. $P(S \mid T) = 0.60$**, and the events are not independent because** $P(S \mid T) eq P(S)$**.** ✓

b. $P(S \mid T) = 0.60$, and the events are independent because $P(S \mid T) > P(S)$.

c. $P(S \mid T) = 0.24$, and the events are not independent because $P(S \mid T) eq P(S)$.

d. $P(S \mid T) = 0.40$, and the events are independent because $P(S \mid T) = P(T)$.

1. **Calculate the conditional probability $P(S \mid T)$:** The condition restricts the sample space to students in Grade 10 (row T), which contains a total of 80 students. Among these, 48 participate in sports: $P(S \mid T) = \frac{n(S \cap T)}{n(T)} = \frac{48}{80} = 0.60$ 2. **Calculate the unconditional probability $P(S)$:** Across all 200 surveyed students, 90 participate in sports: $P(S) = \frac{n(S)}{N} = \frac{90}{200} = 0.45$ 3. **Test for statistical independence:** Two events A and B are statistica.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Transformations & Symmetry

A bag contains 5 red marbles and 4 blue marbles. Two marbles are drawn from the bag without replacement. What is the probability that the first marble drawn is red and the second marble drawn is blue?

a. $\frac{5}{18}$ ✓

b. $\frac{4}{18}$

c. $\frac{20}{81}$

d. $\frac{5}{9}$

This problem involves dependent events, as the marbles are drawn without replacement. We can solve this using a multi-stage probability approach. Step 1: Calculate the probability of drawing a red marble first. There are 5 red marbles and a total of 9 marbles (5 red + 4 blue). So, the probability of drawing a red marble first is: P(1st is Red) = Number of red marbles / Total number of marbles = $\frac{5}{9}$ Step 2: Calculate the probability of drawing a blue marble second, given that the first marble was red. Afte.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Foundations & Reasoning

A point is chosen uniformly at random on the line segment from $x = 0$ to $x = 20$. Let event A be the event that the point lands on the interval $[2, 10]$, and let event B be the event that the point lands on the interval $[6, 16]$. What is the probability that the point lands in A or B, $P(A \cup B)$?

a. $\frac{7}{10}$ ✓

b. $\frac{9}{10}$

c. $\frac{1}{5}$

d. $\frac{1}{2}$

The entire sample space is the segment $[0, 20]$, which has length $20 - 0 = 20$. 1. Calculate $P(A)$: Event A corresponds to $[2, 10]$, with length $10 - 2 = 8$. So, $P(A) = \frac{8}{20}$. 2. Calculate $P(B)$: Event B corresponds to $[6, 16]$, with length $16 - 6 = 10$. So, $P(B) = \frac{10}{20}$. 3. Calculate $P(A \cap B)$: The intersection $A \cap B$ is the interval $[6, 10]$, with length $10 - 6 = 4$. So, $P(A \cap B) = \frac{4}{20}$. Using the Addition Rule of Probability.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Coordinate Geometry

A bag contains 18 marbles, of which 12 are red. If one marble is drawn at random, what is the probability it is red?

a. $\frac{1}{3}$

b. $\frac{2}{3}$ ✓

c. $\frac{3}{2}$

d. $\frac{12}{19}$

$P(\text{red}) = \frac{\text{favorable}}{\text{total}} = \frac{12}{18}$, which simplifies to $\frac{2}{3}$.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Lines & Angles

Two secants are drawn from an external point to a circle, intercepting arcs of 100° and 40°. What is the measure of the angle formed by the secants?

a. 100°

b. **30° ✓**

c. 70°

d. 40°

Angle formed by two secants from an external point $= \frac{1}{2}(\text{far arc} - \text{near arc}) = \frac{1}{2}(100 - 40) = 30^\circ$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Triangles & Congruence

Volume of a rectangular prism with dimensions $2 \times 3 \times 4$:

a. 21

b. 9

c. **24 ✓**

d. 26

Applying Area, Surface Area & Volume principles, $V = lwh = 2 \times 3 \times 4 = 24$. Therefore, the verified correct option is 24.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

In a triangle, the two remote interior angles measure 14° and 10°. What is the measure of the exterior angle at the third vertex?

a. **24° ✓**

b. 14°

c. 156°

d. 10°

By the Exterior Angle Theorem, the exterior angle equals the sum of the two remote interior angles: $14 + 10 = 24^\circ$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

Two chords intersect inside a circle, intercepting arcs of 30° and 178° (the arc and its vertical arc). What is the measure of the angle formed at the intersection?

a. 76°

b. 208°

c. 30°

d. **104° ✓**

Angle formed by two intersecting chords $= \frac{1}{2}(\text{arc}_1 + \text{arc}_2) = \frac{1}{2}(30 + 178) = 104^\circ$.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A rectangle has length 11 and width 6. What is its area?

a. 34

b. **66** ✓

c. 72

d. 17

Applying Area & Perimeter principles, Area $= l \times w = 11 \times 6 = 66$. Therefore, the verified correct option is 66.

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